

Customer Churn Prediction



Predicting Customer Churn and Improving Retention in Telecom using Sas Viya

25FL-CS665-152W – ANALYTIC
TECHNIQUES

FINAL PROJECT

PROFESSOR DESAI

PRESENTER

- KEDEESH NOLAN



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Executive Summary

- Telecommunication companies often struggle to identify customers who are at risk of leaving, leading to lost revenue and higher acquisition costs.
- This project uses the Telco Customer Churn dataset and SAS Viya to analyze service, billing, and demographic patterns that contribute to customer turnover.
- After preparing and cleaning the data, two predictive models were developed (Logistic Regression and Decision Tree) and evaluated for accuracy and interpretability. Logistic Regression emerged as the strongest model and provides a reliable framework for predicting churn.
- The results offer actionable insights that can support retention strategies and improve long-term customer value.



Problem Statement

- Telecommunication companies face high customer churn, which increases operational costs and reduces long-term revenue.
- The organization lacks a clear understanding of which customer characteristics and billing behaviors signal a higher risk of churn.
- Without a data-driven framework, retention strategies are reactive instead of proactive.
- This project aims to identify the key predictors of churn and build a model that can accurately classify customers who are likely to leave.

Data Source & Preparation

- Dataset: Telco Customer Churn (7,043 customers, 21 variables)
- Contains customer demographics, service subscriptions, account details, and billing information
- Target variable: Churn (“Yes” or “No”)
- Key features include:
 - Contract type, Tenure, InternetService, MonthlyCharges, TotalCharges
 - Customer household info (Partner, Dependents)
 - Service add-ons (OnlineSecurity, TechSupport, Streaming)
- Imported into SAS Studio using the Import Wizard
- Raw data issues identified:
 - TotalCharges stored as character with blanks for missing values
 - Mix of numeric + categorical fields
 - CustomerID used only as identifier, not predictive
- Cleaning steps performed:
 - Converted TotalCharges to numeric
 - Replaced blank strings with missing values
 - Dropped/renamed corrected fields
- Final cleaned dataset saved as TELCO_CLEAN and prepared for modeling



Raw Telco Customer Churn Dataset

SAS Studio compute context

New Options View Open Save All

Explorer

- My Favorites
 - CleanTelcoDataset.sas
- Folder Shortcuts
 - Files
 - Home
 - casuser
 - customer.parquet
 - country_lookup.csv
 - home_equity.sashdat
 - TelCO_CLEAN.csv
 - TelCO_CLEAN.sashdat
 - Courses
 - quick-start
 - _Load_Quick_Start_Data.sas
 - CleanTelcoDataset.sas
 - Compute_CAS_CASL.sas
 - country_lookup.csv
 - Create_Email_Message.txt
 - customer.csv
 - customer.sas7bdnt
 - Finance Information.egg
 - Getting Started with PROC Python...
 - Getting Started with the Python E...
 - home_equity.csv
 - home_equity.sas7bdnt

WA_Fn-UseC_Telco-Customer-Churn.csv * WA_Fn-UseC_Telco-Customer-Churn.cti CleanTelcoDataset.sas WORK.TELCO_CLEAN +

Copy to My Snippets Import

	omerID,gender,SeniorCitizen,Partner,Dependents,tenure,PhoneService,MultipleLines,InternetService,OnlineSecurity,OnlineBackup,DevicePro
1	-VHVEG,Female,0,Yes,No,1,No,No phone service,DSL,No,Yes,No,No,No,Month-to-month,Yes,Electronic check,29.85,29.85,No
2	-GNVDE,Male,0,0,No,34,Yes,No,DSL,Yes,No,Yes,No,No,One year,No,Mailed check,56.95,1889.5,No
3	-QPYBK,Male,0,0,No,2,Yes,DSL,Yes,No,No,No,Month-to-month,Yes,Mailed check,53.85,108.15,Yes
4	-CPFCW,Male,0,0,No,45,No,No phone service,DSL,Yes,No,Yes,Yes,No,One year,No,Bank transfer (automatic),42.3,1840.75,No
5	-HQITU,Female,0,0,No,2,Yes,No,Fiber optic,No,No,No,Month-to-month,Yes,Electronic check,70.7,151.65,Yes
6	-CDSCF,Female,0,0,No,8,Yes,Yes,Fiber optic,No,No,Yes,No,Yes,Yes,Month-to-month,Yes,Electronic check,99.65,820.5,Yes
7	-KIOVK,Male,0,0,No,22,Yes,Yes,Fiber optic,No,Yes,No,No,Month-to-month,Yes,Credit card (automatic),89.1,1949.4,No
8	-OKOMC,Female,0,0,No,10,No,No phone service,DSL,Yes,No,No,No,Month-to-month,No,Mailed check,29.75,301.9,No
9	-POOXP,Female,0,Yes,No,28,Yes,Yes,Fiber optic,No,No,Yes,Yes,Yes,Yes,Month-to-month,Yes,Electronic check,104.8,3046.05,Yes
10	-TABGU,Male,0,0,No,62,Yes,No,DSL,Yes,Yes,No,No,No,One year,No,Bank transfer (automatic),56.15,3487.95,No
11	-GRSKD,Male,0,Yes,13,Yes,No,DSL,Yes,No,No,No,Month-to-month,Yes,Mailed check,49.95,587.45,No
12	-LKBCI,Male,0,0,No,16,Yes,No,No internet service,No internet service,No internet service,No internet service,No internet service,N
13	-TTVAX,Male,0,Yes,No,58,Yes,Yes,Fiber optic,No,No,Yes,No,Yes,One year,No,Credit card (automatic),100.35,5681.1,No
14	-XJGEK,Male,0,0,No,49,Yes,Yes,Fiber optic,No,Yes,Yes,No,Yes,Yes,Month-to-month,Yes,Bank transfer (automatic),103.7,5036.3,Yes
15	-JLPIS,Male,0,0,No,25,Yes,No,Fiber optic,Yes,No,Yes,Yes,Yes,Month-to-month,Yes,Electronic check,105.5,2686.05,No
16	-SNQYZ,Female,0,Yes,Yes,69,Yes,Yes,Fiber optic,Yes,Yes,Yes,Yes,Two year,No,Credit card (automatic),113.25,7895.15,No
17	-XMSZG,Female,0,0,No,52,Yes,No,No internet service,No internet service,No internet service,No internet service,No internet service
18	-WOFTT,Male,0,0,No,71,Yes,Yes,Fiber optic,Yes,No,Yes,No,Yes,Two year,No,Bank transfer (automatic),106.7,7382.25,No
19	-MFLUW,Female,0,Yes,Yes,10,Yes,No,DSL,No,No,Yes,No,Month-to-month,No,Credit card (automatic),55.2,528.35,Yes
20	-MYFRB,Female,0,0,No,21,Yes,No,Fiber optic,No,Yes,Yes,No,No,Yes,Month-to-month,Yes,Electronic check,90.05,1862.9,No
21	-QRDMV,Male,1,0,No,1,No,No phone service,DSL,No,No,Yes,No,No,Yes,Month-to-month,Yes,Electronic check,39.65,39.65,Yes
22	-VDCCW,Male,0,Yes,No,12,Yes,No,No internet service,No internet service,No internet service,No internet service,No internet service,
23	-JKSGK,Male,0,0,No,1,Yes,No,No internet service,No internet service,No internet service,No internet service,No internet service,No
24	-WEABW,Female,0,Yes,No,58,Yes,Yes,DSL,No,Yes,No,No,Two year,Yes,Credit card (automatic),59.9,3505.1,No
25	-HRPPA,Male,0,Yes,Yes,49,Yes,No,DSL,Yes,No,Yes,No,Month-to-month,No,Credit card (automatic),59.6,2970.3,No
26	-JENKO,Female,0,0,No,30,Yes,No,DSL,Yes,Yes,No,No,Month-to-month,Yes,Bank transfer (automatic),55.3,1530.6,No
27	-CHFZH,Male,0,Yes,Yes,47,Yes,Yes,Fiber optic,No,Yes,No,Yes,Month-to-month,Yes,Electronic check,99.35,4749.15,Yes
28	-UTDHZ,Male,0,Yes,Yes,1,No,No phone service,DSL,No,Yes,No,No,No,Month-to-month,No,Electronic check,30.2,30.2,Yes
29	-YGLJN,Male,0,Yes,No,72,Yes,Yes,DSL,Yes,Yes,Yes,Yes,Two year,Yes,Credit card (automatic),90.25,6369.45,No
30	-HHUOZ,Female,0,0,Yes,17,Yes,No,DSL,No,No,No,Yes,Month-to-month,Yes,Mailed check,64.7,1093.1,Yes
31	-NFECX,Female,1,Yes,No,71,Yes,Yes,Fiber optic,Yes,Yes,Yes,Yes,No,Two year,Yes,Credit card (automatic),96.35,6766.95,No
32	-XIHWV,Male,1,Yes,No,2,Yes,No,Fiber optic,No,No,Yes,No,Yes,Month-to-month,Yes,Credit card (automatic),95.5,181.65,No
33	-IEAUQ,Female,0,Yes,Yes,27,Yes,No,DSL,Yes,Yes,Yes,No,One year,No,Mailed check,66.15,1874.45,No
34	-EGVHZ,Male,0,0,No,1,Yes,No,No internet service,No internet service,No internet service,No internet service,No internet service,Nc
35	-BMNEE,Male,1,0,No,1,Yes,No,DSL,No,No,No,No,Month-to-month,No,Bank transfer (automatic),45.25,45.25,No
36	
37	

File: /export/viya/homes/knolan4673@monroev.edu/quick-start/WA_Fn-UseC_Telco-Customer-Churn.csv Line 1 Column 1 UTF-8 Recover (2) Submission (0)

Cleaning the Dataset Using SAS Code

The screenshot displays the SAS Studio interface with the following components:

- Explorer:** Shows a file tree with folders like 'My Favorites', 'Folder Shortcuts', and 'Files'. The 'Files' folder contains a sub-folder 'customer' with files like 'customer.parquet', 'country_lookup.csv', 'home_equity.sashdat', 'TELCO_CLEAN.csv', and 'TELCO_CLEAN.sashdat'.
- Code Editor:** Contains SAS code for cleaning the dataset. The code includes comments, data step creation, variable fixing, and a proc contents statement to view the structure of the cleaned dataset.
- Log/Results/Output Data:** The 'Output Data' tab is active, showing a table with 21 columns and 14 rows. The table is titled 'TELCO_CLEAN' and contains customer data.

SAS Code:

```
1 /* Create a cleaned version of the dataset */
2 data work.telco_clean;
3   set work.telco_churn;
4
5   /* Fix TotalCharges variable */
6   /* Replace blanks with missing and convert to numeric */
7   if strip(TotalCharges) = "" then TotalCharges_num = .;
8   else TotalCharges_num = input(TotalCharges, 8.);
9
10  /* Drop the original text version and rename the numeric
11  drop TotalCharges;
12  rename TotalCharges_num = TotalCharges;
13 run;
14
15 /* View structure of cleaned dataset */
16 proc contents data=work.telco_clean; run;
17
```

Output Data (1):

	customerID	gen...	SeniorCitizen	P...
1	7590-VHVEG	Female	0	Yes
2	5575-GNVDE	Male	0	No
3	3668-QPYBK	Male	0	No
4	7795-CFOCW	Male	0	No
5	9237-HQITU	Female	0	No
6	9305-CDSKC	Female	0	No
7	1452-KIOVK	Male	0	No
8	6713-OKOMC	Female	0	No
9	7892-POOKP	Female	0	Yes
10	6388-TABGU	Male	0	No
11	9763-GRSKD	Male	0	Yes
12	7469-LKBCI	Male	0	No
13	8091-TTVAX	Male	0	Yes
14	0280-XJGEX	Male	0	No

Final Cleaned Dataset (TELCO_CLEAN)

The screenshot displays a data analysis application with the following components:

- File Explorer (Left):** Shows a project structure with folders like 'My Favorites', 'Folder Shortcuts', and 'Files'. The 'Files' folder contains various data files including 'customer.parquet', 'country_lookup.csv', 'home_equity.sashdat', 'TELCO_CLEAN.csv', and 'TELCO_CLEAN.sashdat'.
- Main Table (Center):** Displays the 'TELCO_CLEAN' dataset with 7043 rows and 21 columns. The visible columns are: customerID, gender, SeniorCitizen, Partner, Dependents, tenure, and PhoneService. The table shows rows 1 through 17.
- User Menu (Right):** A dropdown menu for the user 'knolan4673@monroe.edu' with options: Recent items, What's new, Settings, About, and Sign out.

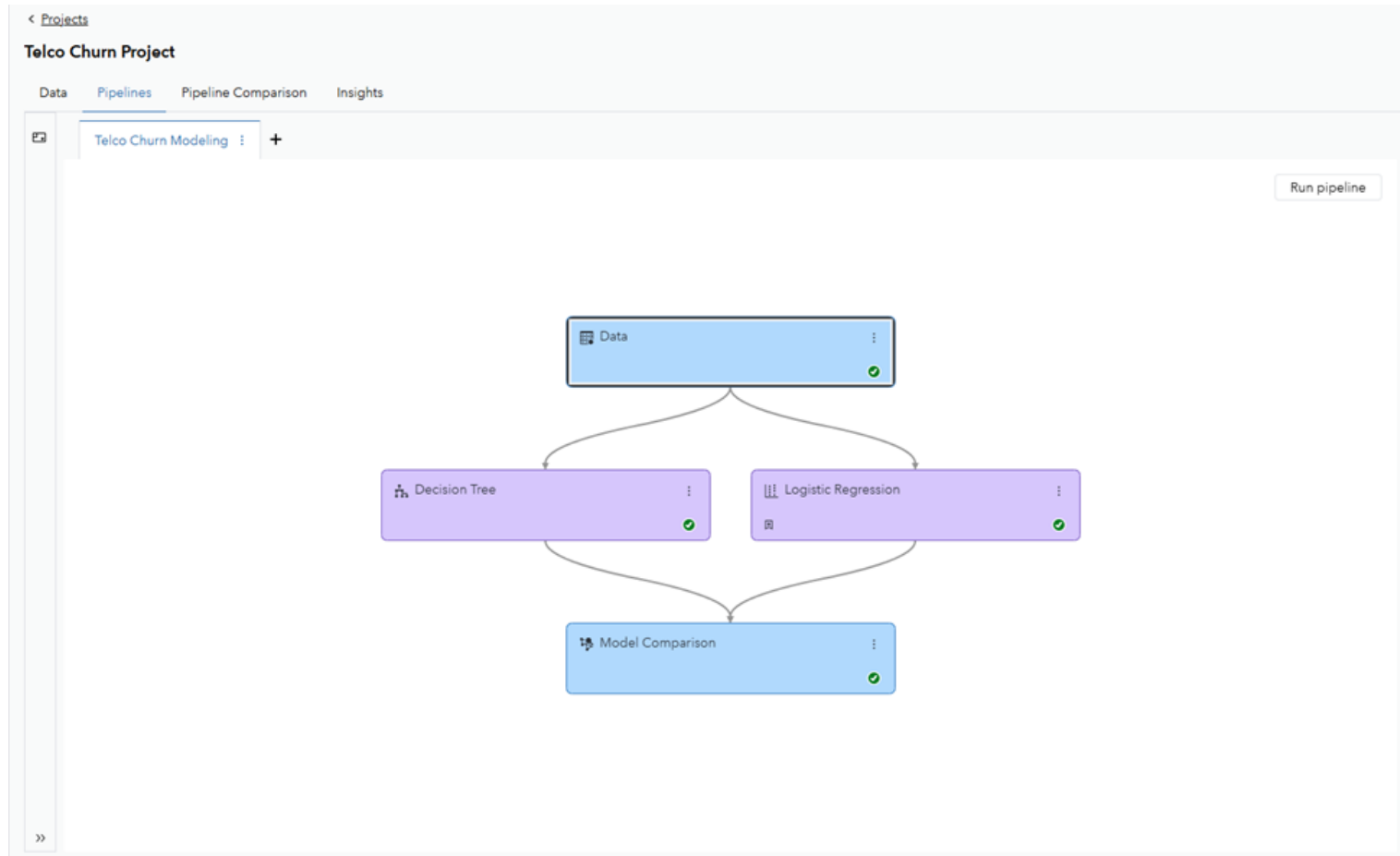
	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService
1	7590-VHVEG	Female	0	Yes	No	1	No
2	5575-GNVDE	Male	0	No	No	34	Yes
3	3668-QPYBK	Male	0	No	No	2	Yes
4	7795-CFOCW	Male	0	No	No	45	No
5	9237-HQITU	Female	0	No	No	2	Yes
6	9305-CDSKC	Female	0	No	No	8	Yes
7	1452-KIOVK	Male	0	No	Yes	22	Yes
8	6713-OKOMC	Female	0	No	No	10	No
9	7892-POOKP	Female	0	Yes	No	28	Yes
10	6388-TABGU	Male	0	No	Yes	62	Yes
11	9763-GRSKD	Male	0	Yes	Yes	13	Yes
12	7469-LKBCI	Male	0	No	No	16	Yes
13	8091-TTVAX	Male	0	Yes	No	58	Yes
14	0280-XJGEX	Male	0	No	No	49	Yes
15	5129-JLPIS	Male	0	No	No	25	Yes
16	3655-SNQYZ	Female	0	Yes	Yes	69	Yes
17	8191-XWSZG	Female	0	No	No	52	Yes

Methodology & Analytics Techniques

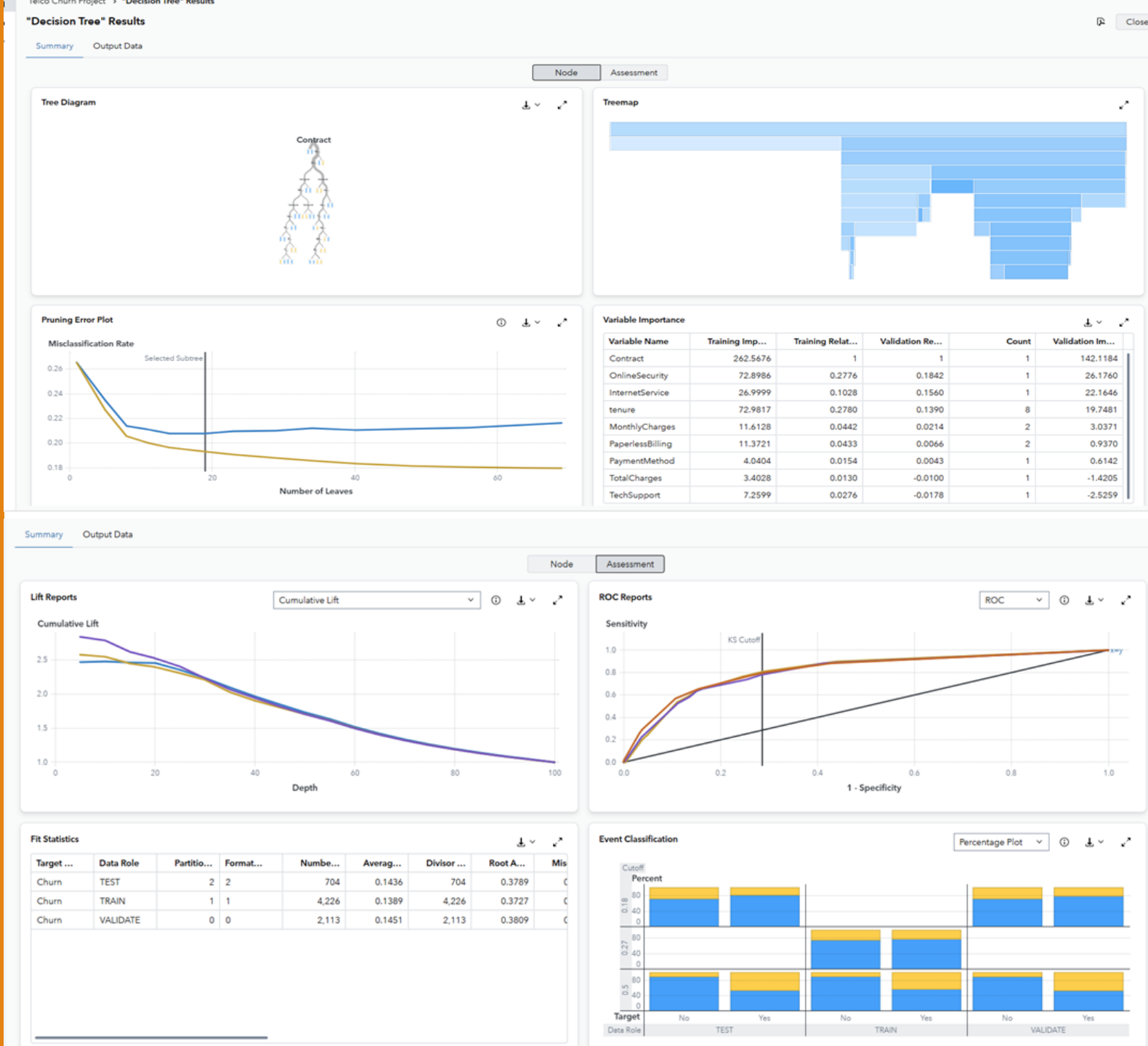
- Overall Approach
 - Followed a structured analytics workflow using SAS Viya.
 - Imported, cleaned, and prepared the Telco Customer Churn dataset for modeling.
 - Conducted exploratory review to understand variable types and data quality.
- Modeling Techniques Used:
 - Logistic Regression
 - Selected for its accuracy and interpretability.
 - Used to estimate churn probability based on customer attributes.
 - Decision Tree
 - Chosen for its ability to capture nonlinear relationships.
 - Useful for identifying key customer segments at high risk of churn.
- Evaluation Process
 - Compared both models using SAS Viya's Model Comparison node.
 - Key evaluation metric: KS (Youden) statistic.
 - Logistic Regression emerged as the champion model.
- Tools & Environment
 - SAS Studio (data cleaning + code execution)
 - SAS Viya Pipelines (model building, training, and comparison)



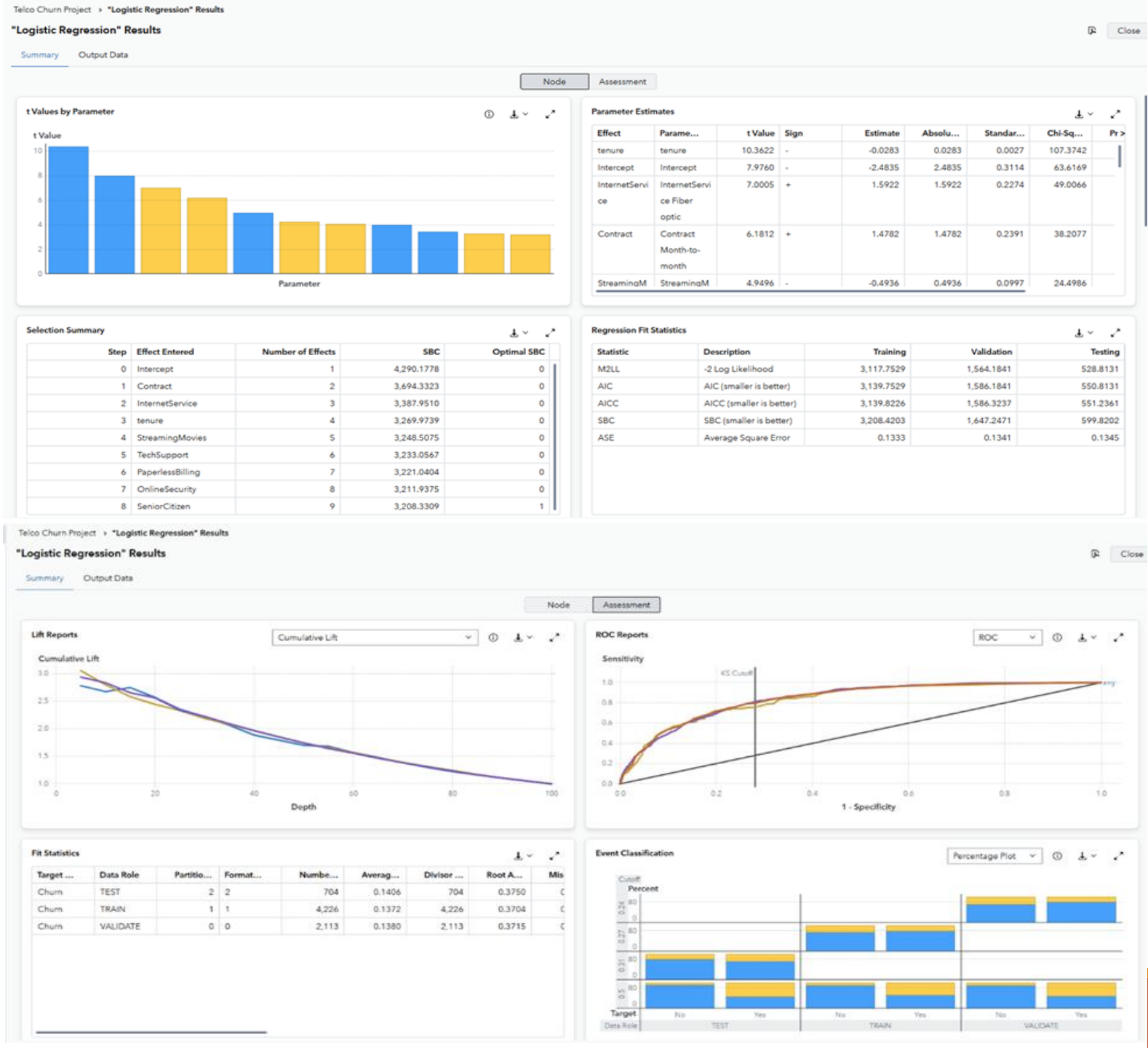
Modeling Workflow in SAS Viya



Decision Tree Model



Logistic Regression Model

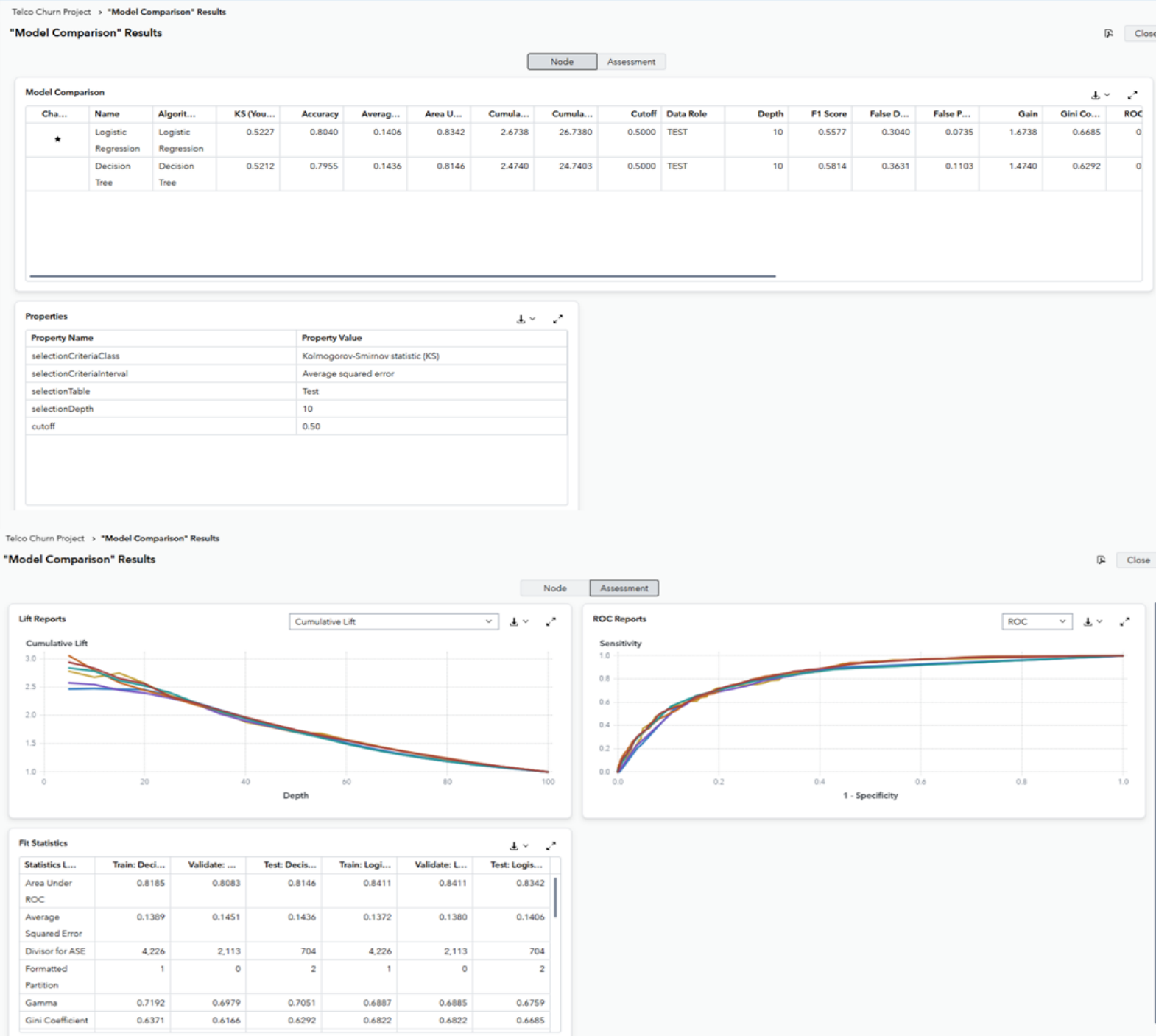


Results & Interpretation

- Logistic Regression outperformed the Decision Tree based on KS (Youden) score and overall model stability.
- The champion model correctly classified 80.4% of customers in the test dataset.
- Top predictors of churn identified by the model include:
 - InternetService
 - OnlineSecurity
 - Tenure
 - TechSupport
 - Contract type
- These factors highlight that customers with shorter tenure, month-to-month contracts, and limited security/tech support services are more likely to churn.
- Insights from the model help the business detect at-risk customers early and tailor retention interventions.



Model Comparison





Business Impact

- The predictive churn model enables the telecom company to identify at-risk customers before they leave, reducing revenue loss.
- Insights from key drivers (e.g., tenure, contract type, security add-ons) allow the business to design targeted retention campaigns.
- Early intervention strategies such as discounts, service upgrades, or personalized outreach which can increase customer loyalty and lifetime value.
- Improved understanding of customer behavior helps optimize pricing, contract structures, and support offerings.
- Provides a simple modeling framework that can be expanded into real operational churn-prevention tools



Personal Touch

- This project allowed me to go beyond the standard workflow by exploring how predictive modeling can directly support customer retention in a real-world telecom environment. I added visual model comparisons, variable importance insights, and pipeline-based modeling in SAS Viya to better understand how different algorithms behave with the same dataset.
- I also incorporated additional screenshots, dataset exploration steps, and a structured cleaning process to demonstrate the full analytical journey from raw data to final model selection. These enhancements showcase not only the technical results, but also my personal approach to organizing, analyzing, and presenting data in a clear and impactful way.

Learnings

Technology Lessons

- Learned how to clean and preprocess the Telco churn dataset, including fixing the TotalCharges variable and creating a cleaned dataset in SAS.
- Gained experience running Decision Tree and Logistic Regression models through SAS Studio pipelines.
- Learned how to compare model outputs by reviewing accuracy, misclassification, and the pipeline's model comparison results.

Management Lessons

- Saw how each phase connects by moving from data understanding to data preparation and then to modeling.
- Learned how important a clean pipeline setup is for keeping the project organized, especially with multiple datasets and scripts in SAS.
- Recognized the need to document preprocessing steps clearly since they directly affect the model pipeline.

Personal Lessons

- Became more comfortable navigating SAS Viya, importing files, cleaning datasets, and running pipelines.
- Gained confidence interpreting the outputs from Decision Tree and Logistic Regression instead of just running them.
- Improved my ability to structure a project from raw CSV to a cleaned dataset to the modeling pipeline the model comparison.

Customer Churn Prediction



THANK
YOU!
